

# **WEEK 3 STATISTICAL ANALYSIS REPORT**

## **Student Performance Analysis**

Student: Geerthana L.

Dataset: StudentsPerformance.csv

Dataset size: 1,000 students, 8 variables

Environment: Google Colab / Python 3

## Executive Summary

This report presents a statistical investigation of the Students Performance dataset containing 1,000 student records. The study examines whether test preparation, lunch type, and parental level of education are associated with student academic performance. Multiple complementary statistical methods were used, including an independent-samples t-test, Cohen's d, chi-square test, one-way ANOVA, Tukey HSD, Levene's test, a 95% confidence interval, normality diagnostics, and linear regression.

The t-test found a statistically significant difference between students who completed the test preparation course and those who did not ( $t = 8.5945$ ,  $p = 4.4267 \times 10^{-17}$ ). Cohen's  $d = 0.5535$  indicates a medium effect. The completed group had a mean average score of 72.67 compared with 65.04 for the no-preparation group, a difference of 7.63 points or approximately 11.73%.

The chi-square test found a significant association between lunch type and performance category ( $\chi^2 = 63.3644$ ,  $df = 2$ ,  $p = 1.7402 \times 10^{-14}$ ). One-way ANOVA found significant differences across parental-education groups ( $F = 10.7531$ ,  $p = 4.3810 \times 10^{-10}$ ). Linear regression showed a strong positive relationship between mathematics and reading scores ( $r = 0.8176$ ,  $R^2 = 0.6684$ ,  $p = 1.7878 \times 10^{-241}$ ).

Overall, the results provide strong evidence of statistical associations among student performance and several educational/contextual factors. Because the dataset is observational, these results should not be interpreted as proof of causation.

## 1. Introduction

Student academic performance can be influenced by educational preparation, learning environment, family background, and contextual conditions. Statistical hypothesis testing provides a rigorous method for determining whether observed differences are likely to represent systematic patterns rather than random variation.

The objective of this Week 3 task is to apply a complete statistical workflow in Python: formulate hypotheses, select appropriate tests, check assumptions, calculate test statistics and p-values, measure effect size, visualize results, and translate statistical findings into practical conclusions.

## 2. Problem Statement

The study investigates whether selected educational and contextual factors are associated with differences in academic performance. The main factors examined are completion of a test preparation course, lunch type, and parental level of education.

## 3. Objectives

- Define clear and testable hypotheses related to student performance.
- Compare average performance between students who completed and did not complete test preparation.
- Examine whether lunch type is associated with performance category.
- Determine whether average performance differs across parental-education groups.
- Measure practical magnitude using Cohen's  $d$ .

- Estimate the overall mean using a 95% confidence interval.
- Check statistical assumptions.
- Quantify the relationship between mathematics and reading scores using linear regression.
- Use Tukey HSD to identify specific parental-education group differences.
- Interpret results without making unsupported causal claims.

## 4. Dataset Description

The Students Performance dataset contains 1,000 student records and eight variables: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score, and writing score.

An average\_score variable was calculated from the mathematics, reading, and writing scores. This composite measure was used as the primary performance measure for group comparisons.

## 5. Research Question and Hypotheses

Research question: Do test preparation, lunch type, and parental level of education show statistically significant relationships with student academic performance?

$H_0$  for the t-test: mean average scores are equal for students who completed and did not complete test preparation.

$H_1$ : mean average scores differ between the two groups.

$H_0$  for the chi-square test: lunch type and performance category are independent.

$H_1$ : lunch type and performance category are associated.

$H_0$  for ANOVA: mean average scores are equal across parental-education groups.

$H_1$ : at least one group mean differs.

$H_0$  for regression: there is no linear relationship between mathematics and reading scores.

$H_1$ : a statistically significant linear relationship exists.

## 6. Methodology

The analysis was conducted in Google Colab using Python. Pandas and NumPy were used for data handling, SciPy for statistical tests, Matplotlib and Seaborn for visualization, and Statsmodels for post-hoc analysis. The significance level was  $\alpha = 0.05$ .

The workflow included descriptive statistics, normality diagnostics, independent-samples t-test, Cohen's d, chi-square test, one-way ANOVA, Tukey HSD, Levene's test, confidence interval estimation, practical significance analysis, and linear regression.

## 7. Descriptive Statistical Summary

Number of students: 1,000

Mean average score: 67.77

Median: 68.33

Standard deviation: 14.26

Minimum: 9.00

Maximum: 100.00

The mean and median are relatively close, while the standard deviation indicates meaningful variation in student performance.

## 8. Normality Analysis

Normality was assessed using formal tests together with a histogram and Q-Q plot. The Q-Q plot shows that most observations follow the reference line through the central range, with larger deviations visible toward the lower tail. Therefore, the distribution is not perfectly normal and the graphical evidence should be considered together with the formal test results.

IMPORTANT: Enter the exact Shapiro-Wilk and D'Agostino p-values from the final Colab output before submission. They are intentionally not invented here because the provided screenshots did not show those numeric values.

## 9. Independent Samples T-Test – Test Preparation

Completed group:  $n = 358$ , mean = 72.67, SD = 13.04

No-preparation group:  $n = 642$ , mean = 65.04, SD = 14.19

t-statistic = 8.5945

p-value =  $4.4267 \times 10^{-17}$

Decision: Reject  $H_0$

The result is statistically significant at  $\alpha = 0.05$ . There is strong evidence that average performance differs between students who completed and did not complete the test preparation course.

## 10. Cohen's d – Effect Size

Cohen's d = 0.5535

Interpretation: Medium effect size

Mean difference = 7.63 points

Percentage improvement = 11.73%

The medium standardized effect indicates that the observed difference is meaningful beyond statistical significance. The completed group scored 7.63 points higher on average.

## 11. Chi-Square Test – Lunch and Performance Category

Contingency table:

Low: Free/Reduced = 64, Standard = 45

Medium: Free/Reduced = 187, Standard = 257

High: Free/Reduced = 104, Standard = 343

Chi-square = 63.3644  
Degrees of freedom = 2  
p-value =  $1.7402 \times 10^{-14}$   
Decision: Reject  $H_0$

There is a statistically significant association between lunch type and performance category. This is an association and should not be interpreted as proof that lunch type itself causes performance differences.

## 12. One-Way ANOVA – Parental Education

F-statistic = 10.7531  
p-value =  $4.3810 \times 10^{-10}$   
Decision: Reject  $H_0$

The ANOVA result indicates that not all parental-education groups have the same mean performance. Because ANOVA identifies an overall difference but not the specific pairs, Tukey HSD was used as a post-hoc analysis.

## 13. Tukey HSD Post-hoc Analysis

Significant reported comparisons included:

- Associate's degree vs High school
- Associate's degree vs Some high school (adjusted p = 0.0183)
- Bachelor's degree vs High school
- Bachelor's degree vs Some high school (adjusted p = 0.0006)
- High school vs Master's degree
- High school vs Some college (adjusted p = 0.0011)
- Master's degree vs Some high school (adjusted p = 0.0007)

These pairwise results explain which parental-education groups contribute to the significant overall ANOVA result.

## 14. Levene's Test

Levene statistic = 2.8851  
p-value = 0.089712  
Decision: Fail to reject  $H_0$

Because  $p > 0.05$ , there is no statistically significant evidence of unequal variances. The equal-variance assumption is therefore supported for the relevant two-group comparison.

## 15. 95% Confidence Interval

Mean average score = 67.77  
95% confidence interval = 66.89 to 68.66

The interval provides a precise estimate of the overall mean performance in this sample of 1,000 students.

## 16. Linear Regression – Mathematics and Reading

Slope = 0.7872

Intercept = 17.1418

$R^2 = 0.6684$

Correlation  $r = 0.8176$

p-value =  $1.7878 \times 10^{-241}$

Regression equation:

Predicted Reading Score =  $17.1418 + 0.7872 \times \text{Math Score}$

The strong positive correlation indicates that students with higher mathematics scores tend to have higher reading scores. Approximately 66.84% of the observed variation in reading scores is explained by mathematics score in this simple model. This relationship is predictive/associational and does not prove causation.

## 17. Practical Significance

Completed preparation mean = 72.67

No preparation mean = 65.04

Mean difference = 7.63 points

Percentage improvement = 11.73%

Practical interpretation = Strong practical improvement

The practical analysis complements the statistical test. The combination of a statistically significant difference and a medium effect size indicates that the result is educationally meaningful within this dataset.

## 18. Consolidated Findings

Independent t-test:  $t = 8.5945$ ,  $p = 4.4267 \times 10^{-17}$  — Significant.

Cohen's d: 0.5535 — Medium effect.

Chi-square:  $\chi^2 = 63.3644$ ,  $p = 1.7402 \times 10^{-14}$  — Significant association.

ANOVA:  $F = 10.7531$ ,  $p = 4.3810 \times 10^{-10}$  — Significant difference.

Levene's test: 2.8851,  $p = 0.089712$  — Equal-variance assumption supported.

95% CI: 66.89–68.66.

Regression:  $R^2 = 0.6684$ ,  $p = 1.7878 \times 10^{-241}$  — Strong significant relationship.

## 19. Discussion

The evidence consistently indicates that student performance is associated with multiple observed factors. The test-preparation comparison is particularly important because it combines statistical significance with a medium practical effect. The chi-square analysis shows a systematic association

between lunch category and performance category, while ANOVA and Tukey HSD demonstrate differences among several parental-education groups.

The strong mathematics-reading relationship indicates that academic outcomes are interconnected across subjects. Together, these results support a multidimensional view of student performance. However, the observational nature of the dataset means that the findings should be interpreted as associations rather than causal effects.

## 20. Educational Implications

- Test-preparation programs may be worth evaluating and strengthening.
- Students with lower performance may benefit from targeted academic support.
- The lunch-performance association suggests that broader contextual and socioeconomic factors deserve attention.
- Differences across parental-education groups indicate the potential importance of access to learning resources.
- The strong math-reading relationship supports integrated academic support across related subjects.

## 21. Limitations

- The dataset is observational, so associations do not establish causation.
- Important variables not included in the dataset may influence performance.
- The composite average gives equal weight to mathematics, reading, and writing.
- Large samples can make formal normality tests sensitive to small deviations.
- Findings should be validated on other student populations before broad generalization.

## 22. Recommendations

- Use multiple regression to examine several predictors simultaneously.
- Test interaction effects to determine whether relationships change across groups.
- Report effect sizes and confidence intervals alongside p-values.
- Validate key findings using another student-performance dataset.
- Use longitudinal or experimental designs for stronger causal conclusions.

## 23. Conclusion

This Week 3 study successfully applied a comprehensive statistical hypothesis-testing workflow to the Students Performance dataset. The independent-samples t-test found a statistically significant difference between test-preparation groups, with Cohen's d indicating a medium effect. The chi-square test identified a significant association between lunch type and performance category. ANOVA and Tukey HSD demonstrated significant differences across selected parental-education groups.

Levene's test supported the equal-variance assumption for the relevant comparison. The overall mean was estimated at 67.77 with a 95% confidence interval of 66.89–68.66. Linear regression revealed a strong positive relationship between mathematics and reading scores, with  $R^2 = 0.6684$ . Overall, the

results show strong statistical evidence of associations between student performance and several educational/contextual factors, while causal claims require further study.

## 24. References

1. SciPy Documentation – Statistical functions and hypothesis testing.
2. Pandas Documentation – Data manipulation and analysis in Python.
3. Matplotlib Documentation – Data visualization in Python.
4. Seaborn Documentation – Statistical data visualization.
5. Statsmodels Documentation – Statistical models and post-hoc analysis.
6. Students Performance dataset – StudentsPerformance.csv.

## 25. Appendix – Core Python Code

Average score:

```
df['total_score'] = df['math score'] + df['reading score'] + df['writing score']
df['average_score'] = df['total_score'] / 3
```

T-test:

```
from scipy.stats import ttest_ind
completed = df[df['test preparation course']=='completed']['average_score']
none = df[df['test preparation course']=='none']['average_score']
t_stat, p_value = ttest_ind(completed, none, equal_var=True)
```

Chi-square:

```
from scipy.stats import chi2_contingency
table = pd.crosstab(df['lunch'], df['performance_category'])
chi2, p, dof, expected = chi2_contingency(table)
```

ANOVA:

```
from scipy.stats import f_oneway
groups = [g['average_score'].values for _, g in df.groupby('parental level of education')]
f_stat, p_value = f_oneway(*groups)
```

Levene:

```
from scipy.stats import levene
stat, p = levene(completed, none)
```

Regression:

```
from scipy.stats import linregress
result = linregress(df['math score'], df['reading score'])
```

Confidence interval:

```
from scipy import stats
mean = df['average_score'].mean()
sem = stats.sem(df['average_score'])
ci = stats.t.interval(0.95, len(df)-1, loc=mean, scale=sem)
```

Normality:

```
from scipy.stats import shapiro, normaltest, probplot
scores = df['average_score'].dropna()
print(shapiro(scores))
print(normaltest(scores))
probplot(scores, dist='norm', plot=plt)
plt.show()
```

## Final Submission Checklist

- ☐ Verify the exact Shapiro-Wilk and D'Agostino p-values from Colab.
- ☐ Insert the main charts from Colab into the report.
- ☐ Keep the complete executable notebook as supporting evidence.
- ☐ Add institution, course, mentor, and submission date if required.
- ☐ Proofread all statistical conclusions before submission.